

palimpsestr: Probabilistic Decomposition of Archaeological Palimpsests Using Stratigraphic Entanglement Fields

Enzo Cocca

2026-04-09

Contents

0.1	Introduction	2
0.2	Methodology	3
0.2.1	The Stratigraphic Entanglement Index (SEI)	3
0.2.2	The Diagonal Gaussian Mixture EM	3
0.2.3	Diagnostic Statistics	4
0.3	The Dataset: Villa Romana	4
0.4	Phase Count Selection	6
0.5	Fitting the Model ($K = 4$)	6
0.5.1	Model Overview	8
0.5.2	EM Convergence	9
0.6	Spatial Diagnostics	9
0.6.1	Phase-field Map	9
0.6.2	Entropy Map	9
0.6.3	Energy Map (ESE)	9
0.6.4	Intrusion Detection Map	14
0.6.5	Vertical Phase Profile	14
0.7	Intrusion Analysis	16
0.8	Stratigraphic Unit Purity	16
0.9	Phase Transition Matrix	18
0.10	Model Validation	18
0.11	Sensitivity Analysis of SEI Weights	19
0.12	Bootstrap Confidence Intervals	20
0.13	Harris Matrix Integration	22
0.13.1	Validate phases against stratigraphy	22
0.14	Interpretive Report	23

0.15	Structured Exports	26
0.16	GIS Export	26
0.17	Interactive Visualization	26
0.18	Comparison with Traditional Approaches	26
0.19	Model Limitations and Methodological Notes	27
0.19.1	Diagonal Covariance Assumption	27
0.19.2	SEI Component Normalisation	27
0.19.3	PDI Interpretation	27
0.19.4	Intrusion Detection	27
0.19.5	Phase Labelling	27
0.19.6	Cross-Validation	27
0.19.7	Scalability	28
0.20	Interactive Shiny Dashboard	28
0.21	Advanced Feature Space Options	28
0.21.1	Chronological Precision Weighting	28
0.21.2	Residuality Detection	28
0.21.3	Class Scaling	28
0.21.4	Taphonomic Score as Feature	29
0.21.5	Sub-class Encoding	29
0.21.6	Using All Five Together	29
0.22	Conclusion	29

*Archaeological palimpsests — deposits in which material from multiple occupation phases is superimposed and partially intermixed — represent one of the most persistent analytical challenges in field archaeology. The **palimpsestr** R package implements the Stratigraphic Entanglement Field (SEF) framework, a probabilistic method for decomposing palimpsests into latent depositional phases by jointly modelling four evidence domains: spatial proximity, vertical distribution, chronological overlap, and cultural similarity. The package provides a diagonal-covariance Gaussian mixture model fitted via Expectation-Maximisation, augmented with taphonomic weighting and stratigraphic entanglement penalties. Three interpretable diagnostics — the Stratigraphic Entanglement Index (SEI), Excavation Stratigraphic Energy (ESE), and Palimpsest Dissolution Index (PDI) — allow practitioners to assess deposit coherence, detect intrusive finds, and evaluate overall phase separability. This article describes the methodology and demonstrates a complete analytical workflow on a realistic multi-period Roman villa dataset.*

0.1 Introduction

Archaeological palimpsests arise when successive episodes of human occupation leave material traces within the same sedimentary matrix, producing deposits in which artefacts, ecofacts, and features from chronologically distinct phases co-occur in close spatial and vertical proximity. The phenomenon is ubiquitous: from deeply stratified Near Eastern tells to shallow open-air Palaeolithic sites where millennia of activity are compressed into a few centimetres of deposit. Disentangling the depositional history of such contexts is essential for any behavioural, technological, or settlement-pattern inference, yet the analytical toolkit available to excavators has remained surprisingly limited.

The Harris Matrix (Harris, 1989) provides a rigorous formalism for recording observed stratigraphic relationships, but its relationships are deterministic and binary — a unit either overlies another or it does not. It cannot express the gradual, probabilistic mixing that characterises many palimpsests. Conversely, spatial clustering methods such as k-means (Kintigh and Ammerman, 1982) operate on coordinate data alone, ignoring vertical stratigraphy, chronological evidence from radiocarbon or ceramic seriation, and cultural-typological information. Neither approach offers a unified probabilistic framework, and neither produces quantitative diagnostics for deposit integrity or mixing intensity.

palimpsestr addresses these limitations by integrating four evidence domains — horizontal coordinates, vertical elevation, chronological range, and cultural class — into a single model that produces soft (probabilistic) phase assignments together with three diagnostic statistics. The Stratigraphic Entanglement Index (SEI) quantifies pairwise depositional coherence; Excavation Stratigraphic Energy (ESE) captures local disruption; and the Palimpsest Dissolution Index (PDI) summarises global phase separability. All three are interpretable by field practitioners and can be exported to GIS for spatial interrogation.

0.2 Methodology

0.2.1 The Stratigraphic Entanglement Index (SEI)

The SEI quantifies how strongly two finds i and j are “entangled” — that is, how much evidence supports their co-occurrence within the same depositional event. It is defined as:

$$SEI_{ij} = \frac{w_s}{d_{xy}(i, j) + \epsilon} + \frac{w_z}{\max(|z_i - z_j|, z_{floor})} + w_t \cdot O_t(i, j) + w_c \cdot \mathbb{1}[c_i = c_j]$$

where:

- $d_{xy}(i, j)$ is the Euclidean distance in the horizontal plane, and ϵ is a small constant preventing division by zero. The first term captures spatial proximity: finds close together in plan are more likely to derive from the same depositional event.
- $|z_i - z_j|$ is the absolute vertical separation, with z_{floor} a minimum threshold. The second term captures stratigraphic proximity: finds at the same depth are more likely contemporaneous.
- $O_t(i, j)$ is the chronological overlap ratio, computed from the intersection of the temporal ranges $[t_{min,i}, t_{max,i}]$ and $[t_{min,j}, t_{max,j}]$ relative to their union. Finds whose date ranges overlap strongly receive higher entanglement scores.
- $\mathbb{1}[c_i = c_j]$ is an indicator function equal to 1 when both finds belong to the same cultural or typological class, and 0 otherwise.
- w_s, w_z, w_t, w_c are domain weights (defaulting to equal weighting) that allow the analyst to encode prior knowledge about which evidence domains are most reliable for a given site.

Each SEI component is normalised to $[0, 1]$ by dividing by its within-dataset maximum, making weights directly interpretable as relative importance. The SEI matrix is row-normalised and used both as a regularisation penalty during model fitting and as the basis for the ESE diagnostic.

0.2.2 The Diagonal Gaussian Mixture EM

Phase decomposition proceeds via a Gaussian mixture model fitted by Expectation-Maximisation (EM). The feature matrix for each find is constructed by concatenating: scaled horizontal coordinates (x, y) , vertical coordinate (z) , temporal midpoint and temporal span of the date range, and a one-hot encoding of the cultural class variable. This multi-domain feature representation ensures that the clustering draws on all available evidence simultaneously.

The algorithm is initialised with K-means, and the resulting hard assignments are converted to softmax probabilities. The EM procedure then iterates between an E-step (computing posterior phase membership probabilities given current parameters) and an M-step (updating mixture weights, means, and diagonal covariance matrices given current memberships). Two modifications distinguish SEF from a standard Gaussian mixture:

1. **Taphonomic weighting:** if a taphonomic integrity score is available for each find, it is used to down-weight poorly preserved or disturbed items during parameter estimation, reducing their influence on cluster centroids.
2. **Stratigraphic entanglement penalties:** the SEI matrix is incorporated as a soft constraint, encouraging finds with high mutual entanglement to receive similar phase assignments.

Convergence is assessed by monitoring the log-likelihood; the algorithm terminates when the relative change falls below a tolerance threshold or a maximum number of iterations is reached.

0.2.3 Diagnostic Statistics

Three statistics summarise different aspects of the decomposition:

- **Excavation Stratigraphic Energy (ESE):** for each find, ESE measures its dissimilarity from its spatial neighbours in terms of phase assignment. A high ESE value indicates that a find is surrounded by items assigned to different phases — a signature of depositional disruption, bioturbation, or post-depositional displacement.
- **Palimpsest Dissolution Index (PDI):** a global statistic defined as $PDI = 1 - \bar{H}/\log(K)$, where $\bar{H}$ is the mean Shannon entropy of the phase-probability vectors across all finds and K is the number of phases. PDI ranges from 0 (complete mixing; every find is equally likely to belong to any phase) to 1 (perfect separation; every find is assigned to exactly one phase with certainty). Values above 0.7 generally indicate well-resolved phases.
- **Entropy:** the Shannon entropy of each find's phase-probability vector, $H_i = -\sum_k p_{ik} \log p_{ik}$. Low entropy means confident assignment; high entropy signals ambiguity, typically at phase boundaries or within heavily mixed zones.
- **Intrusion score:** a composite metric combining high entropy, high ESE, and low SEI connectivity. Finds exceeding a threshold on this composite are flagged as candidate intrusions — items that are spatially or stratigraphically anomalous relative to their assigned phase.

0.3 The Dataset: Villa Romana

We demonstrate the complete analytical workflow using `villa_romana`, a real archaeological dataset from Poggio Gramignano (Lugnano in Teverina, TR), a multi-period Roman villa with an annexed Late Antique infant cemetery. The dataset contains 615 inventoried finds from 54 stratigraphic units, with taphonomic scores assigned by the excavator.

```
library(palimpsestr)
data(villa_romana)
cat("Finds:", nrow(villa_romana), "\n")
#> Finds: 615
cat("Stratigraphic units:", length(unique(villa_romana$context)), "\n")
#> Stratigraphic units: 54
cat("Material classes:", length(unique(villa_romana$class)), "\n")
```

```
#> Material classes: 17
cat("Date range:", min(villa_romana$date_min), "to", max(villa_romana$date_max), "\n")
#> Date range: -650 to 550
cat("Mean taf_score:", round(mean(villa_romana$taf_score), 3), "\n")
#> Mean taf_score: 0.778
```

The site spans eight archaeological periods:

Period	Chronology	Description
10	650–350 BCE	Pre-Roman (foundation pits)
9	30 BCE–80 CE	Augustan–Flavian (villa construction)
8	80–137 CE	Flavian
7	138–179 CE	Antonine–Severan
6	180–324 CE	Severan–Late Antique
5	324–425 CE	Late Antique
4	425–450 CE	Late Antique (cemetery)
3	450–550 CE	Late Antique–Early Medieval

The deposit includes levelling fills (US 76, 117, 118), backfills, and redeposited material, reflected in the taphonomic scores (taf_score: 0.5 for disturbed, 0.7 for redeposited, 1.0 for in situ).

```
cat("Material classes:\n")
#> Material classes:
sort(table(villa_romana$class), decreasing = TRUE)
#>
#>      Reperto Ceramico      Reperto Anforaceo Materiale Da Costruzione
#>           417           76           38
#>      Reperto Vitreo      Reperto Metallico      Reperto Numismatico
#>           25           19           10
#>      Reperti Archeozoologici      Reperto Lapideo      Campionature
#>           7           6           5
#>      Scarti Di Produzione      Intonaco Dipinto      Materiale Lapideo
#>           5           1           1
#>      Pietre Dure-Gemme      Reperti Antropologici      Reperto Faunistico
#>           1           1           1
#>      Resti Di Pavimentazioni      Stucco
#>           1           1
```

```
str(villa_romana)
#> 'data.frame':   615 obs. of  9 variables:
#>  $ id      : chr  "VRPG_0001" "VRPG_0002" "VRPG_0003" "VRPG_0004" ...
#>  $ x       : num  2299041 2299041 2299041 2299041 2299041 ...
#>  $ y       : num  4714126 4714126 4714126 4714126 4714126 ...
#>  $ z       : num  284 284 284 284 284 ...
#>  $ context  : chr  "US_102" "US_102" "US_102" "US_102" ...
#>  $ date_min : num  425 425 425 425 425 425 425 425 425 425 ...
#>  $ date_max : num  450 450 450 450 450 450 450 450 450 450 ...
#>  $ class    : chr  "Reperto Ceramico" "Reperto Ceramico" "Reperto Ceramico" "Reperto Ceramico" ...
#>  $ taf_score: num  0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 ...
#> - attr(*, "source")= chr "Poggio Gramignano (VRPG 3004), pyArchInit database"
#> - attr(*, "taf_source")= chr "Compiled by Roberto Montagnetti (April 2026)"
```

0.4 Phase Count Selection

When the number of phases is unknown, `compare_k()` fits models across a range of K values and reports BIC, ICL, and PDI for each. The optimal K minimises BIC while maximising PDI.

```
ck <- compare_k(
  villa_romana,
  k_values = 2:7,
  tafonomy = "taf_score",
  context = "context",
  seed = 42
)
print(ck)
```

#>	k	pdi	mean_entropy	mean_local_sei	mean_energy	loglik	bic
#>	1 2	1.0000000	5.286895e-11	899.8031	2.005266	36080.89	-71590.26
#>	2 3	1.0000000	1.783246e-12	899.8031	2.005266	43697.29	-86534.09
#>	3 4	0.9999816	2.546414e-05	899.8031	2.005266	45013.08	-88876.68
#>	4 5	0.9999983	2.718478e-06	899.8031	2.005266	49117.20	-96795.95
#>	5 6	0.9999246	1.350777e-04	899.8031	2.005266	55589.89	-109452.37
#>	6 7	0.9999968	6.252360e-06	899.8031	2.005266	56618.23	-111220.06

#>		icl	tot_withinss	pseudo_bic
#>	1	-71590.26	1700.4768	908.0320
#>	2	-86534.09	1580.2767	1004.2228
#>	3	-88876.71	1543.7001	1131.0966
#>	4	-96795.96	1464.8319	1240.1206
#>	5	-109452.54	689.4588	917.9392
#>	6	-111220.07	355.2304	651.3836

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_compare_k(ck)
}
```

0.5 Fitting the Model ($K = 4$)

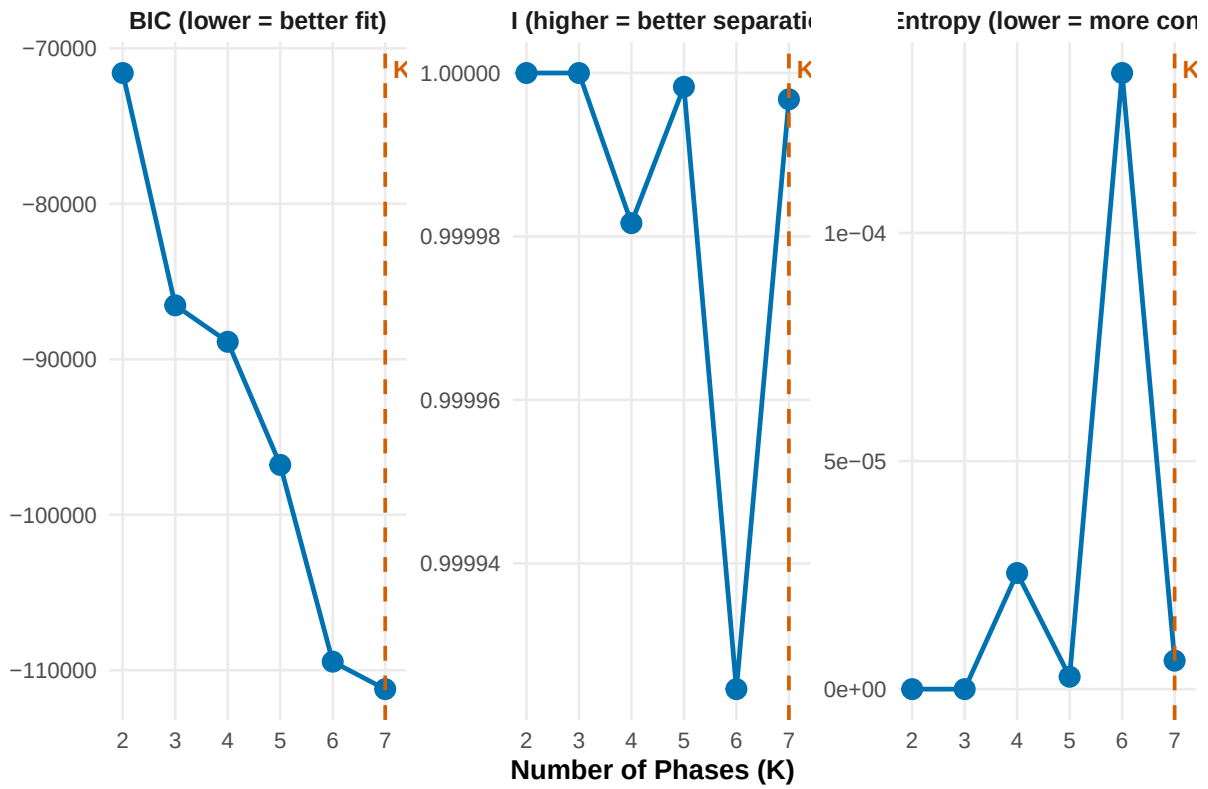
Based on the selection diagnostics and archaeological knowledge of the site (4 occupation periods), we fit with $K = 4$ and multiple random initialisations.

```
fit <- fit_sef(
  villa_romana,
  k = 4,
  tafonomy = "taf_score",
  context = "context",
  n_init = 10,
  seed = 42
)
print(fit)
```

```
#> <sef_fit>
#> Observations: 615
#> Estimated phases: 4
#> Mean entropy: 0.000
#> Mean local SEI: 899.803
#> Mean energy: 2.005
```

Phase Count Selection

Comparing K = 2 to 7 | Optimal K = 7 (minimum BIC)



Generated by palimpsestr 0.12.0

Figure 1: Model selection diagnostics: BIC, PDI, and mean entropy across candidate phase counts. The dashed line marks the optimal K.

```
#> LogLik: 52469.243
#> BIC: -103789.015
#> ICL: -103789.015
#> PDI: 1.000
#> Converged: yes
#> Initialisations: 10
```

```
summary(fit)
#> $n
#> [1] 615
#>
#> $k
#> [1] 4
#>
#> $pdi
#> [1] 1
#>
#> $mean_entropy
#> [1] 3.867719e-10
#>
#> $mean_energy
#> [1] 2.005266
#>
#> $loglik
#> [1] 52469.24
#>
#> $bic
#> [1] -103789
#>
#> $phase_count
#>
#>   1   2   3   4
#> 117 82 25 391
```

0.5.1 Model Overview

```
cat("PDI:", pdi(fit), "\n")
#> PDI: 1
sef_summary(fit)
#> $n
#> [1] 615
#>
#> $k
#> [1] 4
#>
#> $pdi
#> [1] 1
#>
#> $mean_entropy
#> [1] 3.867719e-10
#>
#> $mean_energy
```



```
#> [1] 2.005266
#>
#> $loglik
#> [1] 52469.24
#>
#> $bic
#> [1] -103789
#>
#> $phase_count
#>
#>   1   2   3   4
#> 117  82  25 391
```

0.5.2 EM Convergence

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_convergence(fit)
}
```

0.6 Spatial Diagnostics

0.6.1 Phase-field Map

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_phasefield(fit)
}
```

The phase-field map reveals the spatial structure of the decomposition. The model assigns each find to one of four chronologically coherent phases despite the presence of post-depositional disturbance.

0.6.2 Entropy Map

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_entropy(fit)
}
```

Entropy mapping is particularly useful for identifying the spatial extent of mixing. Zones of elevated entropy correspond to areas where multiple depositional phases are interleaved and cannot be cleanly separated.

0.6.3 Energy Map (ESE)

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_energy(fit)
}
```

The energy map highlights local disruptions. Clusters of high-ESE finds may indicate bioturbation channels, pit cuts, or other post-depositional disturbances.

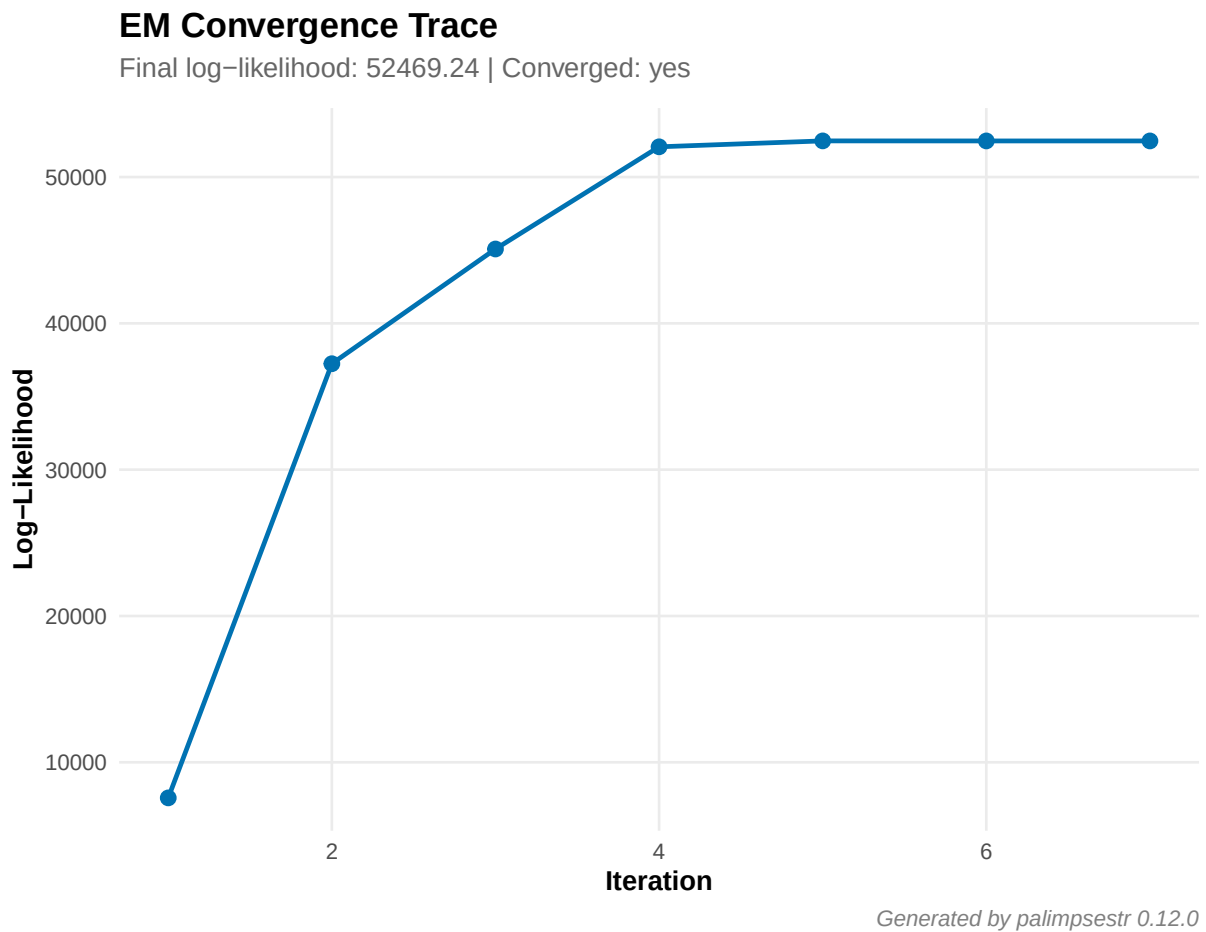


Figure 2: EM convergence trace showing log-likelihood stabilisation across iterations.

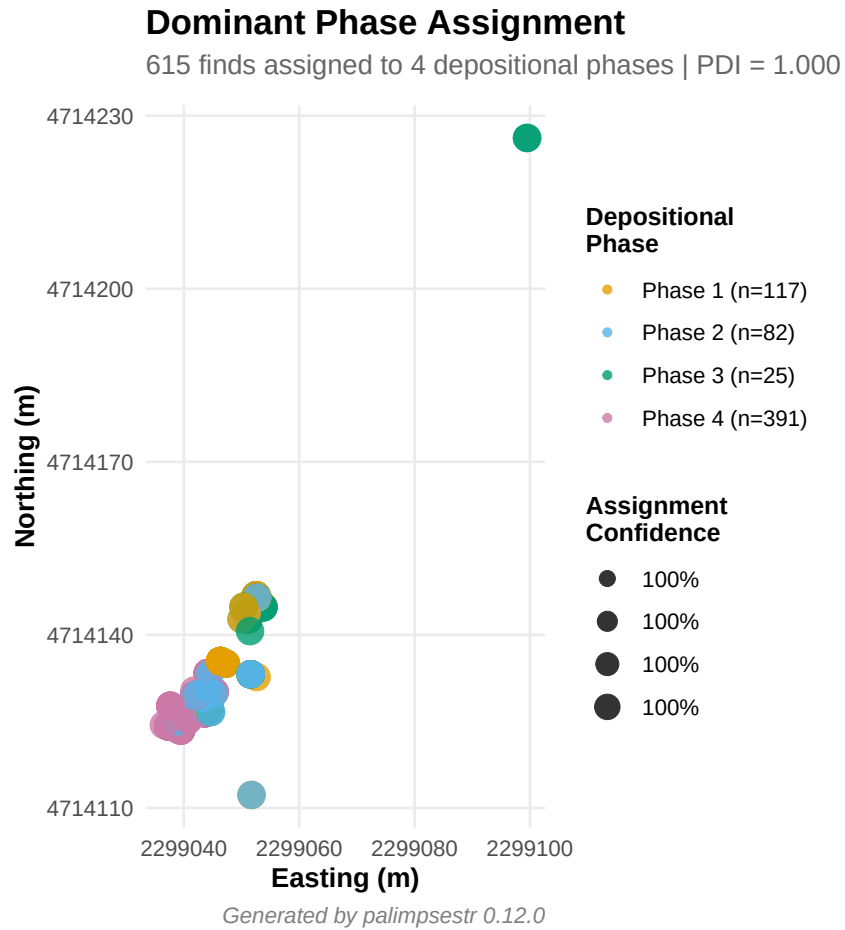


Figure 3: Dominant phase assignment. Point size reflects assignment confidence (inverse entropy). Larger points indicate higher confidence.

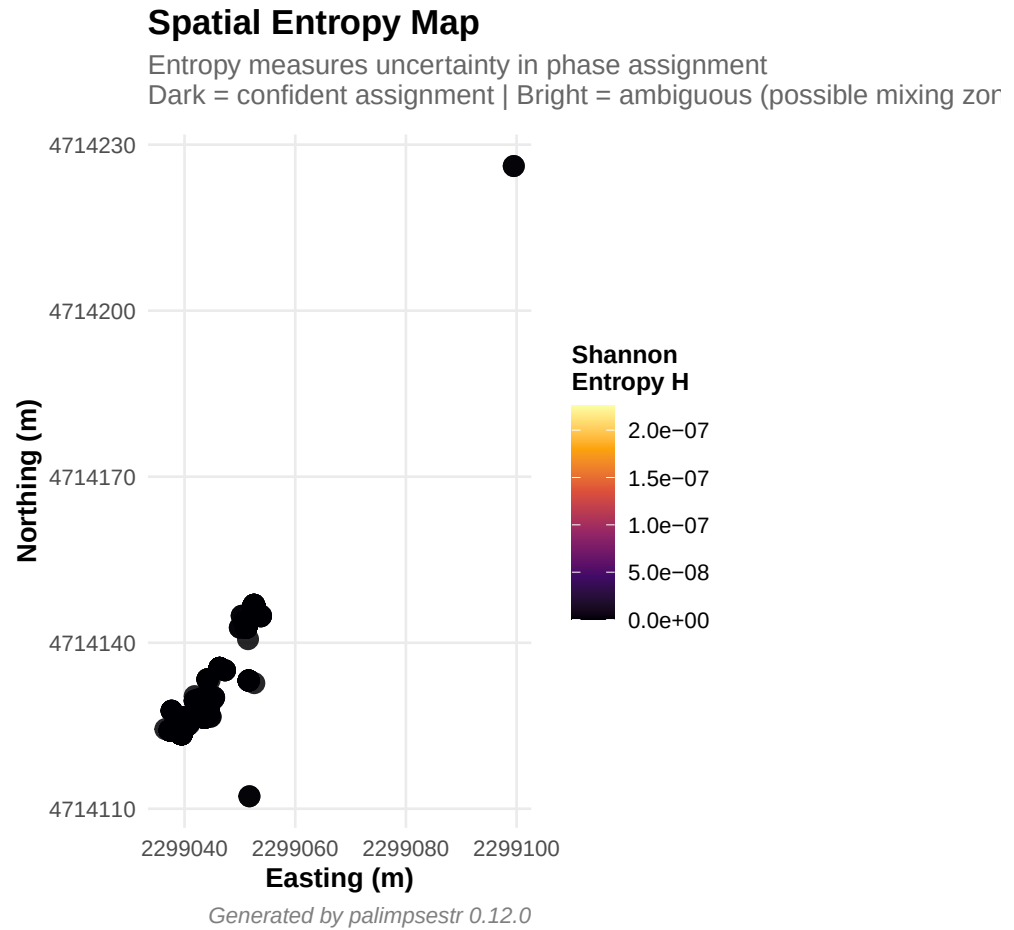
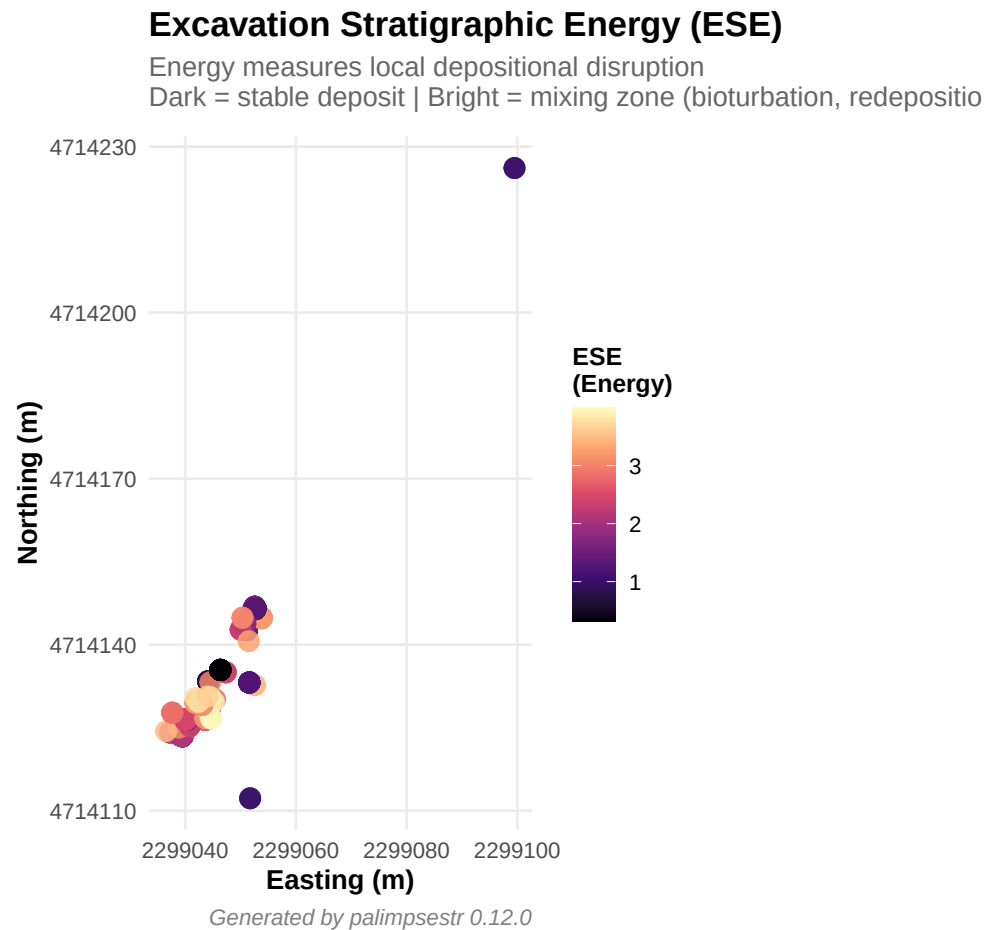


Figure 4: Spatial distribution of Shannon entropy. High-entropy zones (bright) mark areas of uncertain phase assignment, typically at phase boundaries or in heavily mixed deposits.



0.6.4 Intrusion Detection Map

```
if (requireNamespace("ggplot2", quietly = TRUE)) {  
  gg_intrusions(fit, top_n = 10)  
}
```

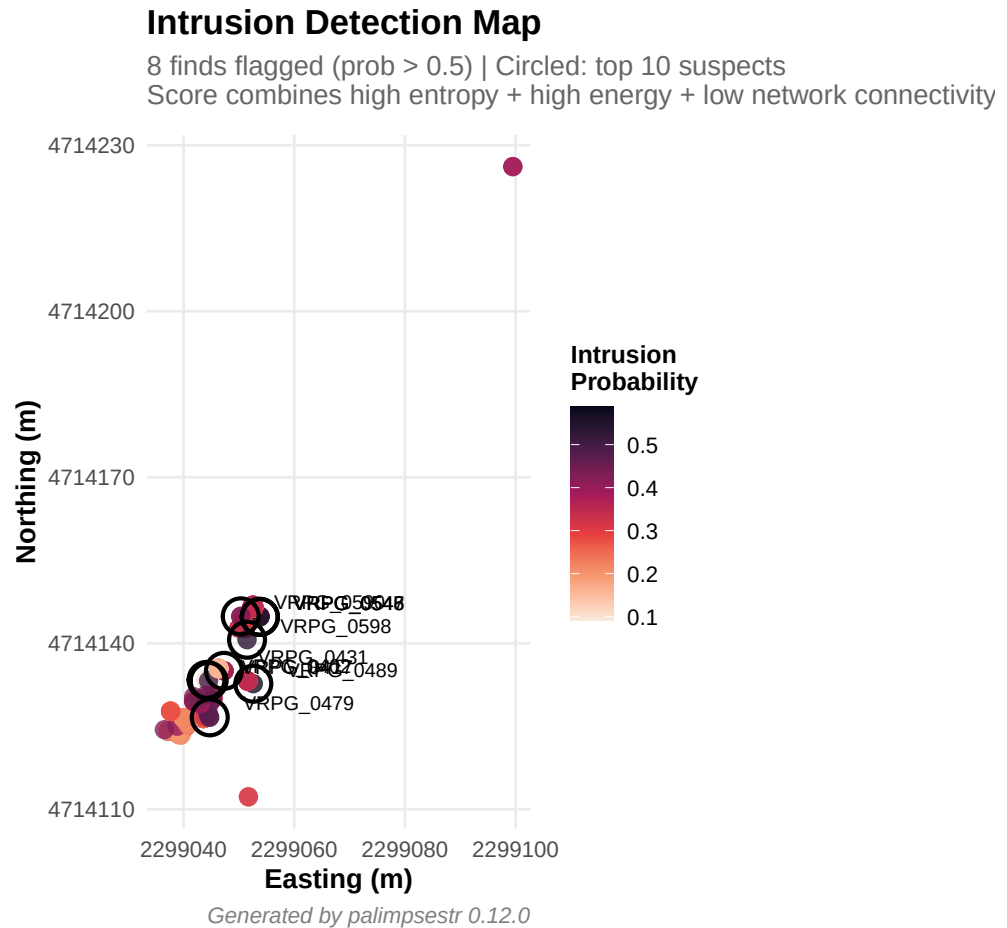


Figure 6: Candidate intrusions overlaid on the phase-field map. Circled finds exhibit high entropy, high stratigraphic energy, and low entanglement with their neighbours. Top 10 suspects are labelled.

0.6.5 Vertical Phase Profile

```
if (requireNamespace("ggplot2", quietly = TRUE)) {  
  gg_phase_profile(fit)  
}
```

The vertical profile confirms the expected stratigraphic ordering: earlier phases occupy deeper strata, later phases are shallower.

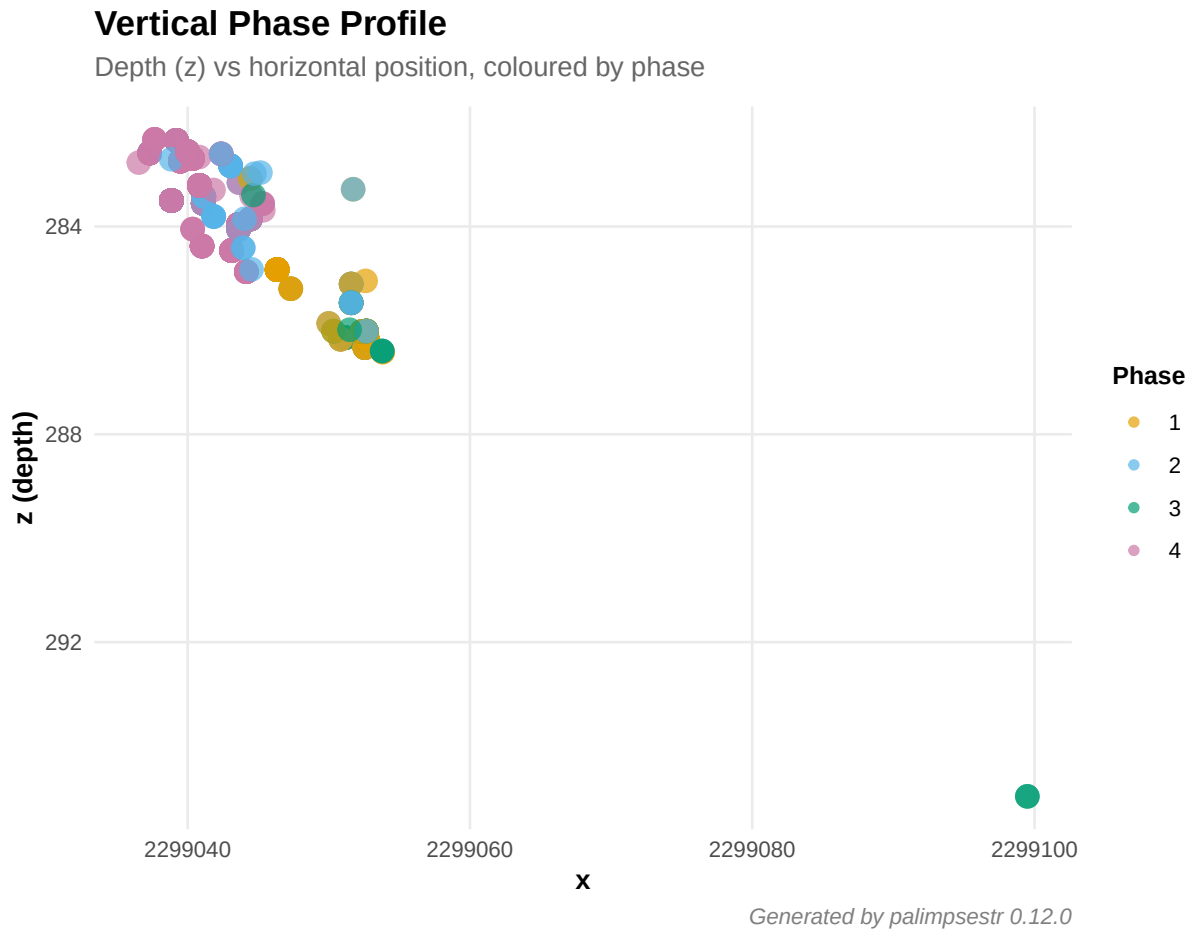


Figure 7: Vertical phase profile: depth vs horizontal position. Phase 1 (deepest) corresponds to the earliest occupation; Phase 4 (shallowest) to the most recent.

0.7 Intrusion Analysis

The `detect_intrusions()` function computes a composite intrusion probability for each find, combining entropy, ESE, and SEI connectivity into a single score.

```
di <- detect_intrusions(fit)
suspects <- di[di$intrusion_prob > 0.5, ]
cat("Suspected intrusions:", nrow(suspects), "out of", nrow(villa_romana), "finds",
    sprintf("(%.1f%%)\n", 100 * nrow(suspects) / nrow(villa_romana)))
#> Suspected intrusions: 8 out of 615 finds (1.3%)

if (nrow(suspects) > 0) {
  # Merge with source data and phase assignments
  phase_tab <- as_phase_table(fit)
  susp_data <- merge(suspects, villa_romana[, c("id", "context", "class")], by = "id")
  susp_data <- merge(susp_data, phase_tab[, c("id", "dominant_phase")], by = "id")
  susp_data <- susp_data[order(susp_data$intrusion_prob, decreasing = TRUE), ]
  susp_show <- susp_data[, c("id", "context", "class", "dominant_phase", "intrusion_prob")]
  names(susp_show) <- c("Find", "US", "Class", "Phase", "Intrusion Prob.")
  knitr::kable(head(susp_show, 15), row.names = FALSE, digits = 3,
                caption = "Top suspected intrusions ranked by intrusion probability.")
}
```

Table 1: Top suspected intrusions ranked by intrusion probability.

Find	US	Class	Phase	Intrusion Prob.
VRPG_0402	US_151	Reperto Anforaceo	4	0.588
VRPG_0489	US_276	Reperto Metallico	1	0.573
VRPG_0431	US_204	Materiale Da Costruzione	2	0.559
VRPG_0590	US_316	Intonaco Dipinto	3	0.539
VRPG_0598	US_324	Reperti Antropologici	3	0.539
VRPG_0548	US_304	Campionature	3	0.538
VRPG_0547	US_304	Scarti Di Produzione	3	0.537
VRPG_0546	US_304	Materiale Da Costruzione	2	0.525

Flagged finds warrant closer examination: they may represent genuinely intrusive material (e.g., items displaced by animal burrowing or construction cuts), or they may sit at genuine phase boundaries where assignment is inherently ambiguous.

0.8 Stratigraphic Unit Purity

The `us_summary_table()` function aggregates diagnostics per stratigraphic unit, reporting dominant phase, purity, mean entropy, energy, and intrusion count.

```
us_tab <- us_summary_table(fit)
knitr::kable(us_tab, row.names = FALSE, digits = 3,
              caption = "Per-US diagnostics: dominant phase, purity, mean entropy, mean ESE, and number of")
```


Table 2: Per-US diagnostics: dominant phase, purity, mean entropy, mean ESE, and number of intrusions.

context	n_finds	dominant_phase	purity	mean_entropy	mean_energy	mean_local_sei	n_intrusions
US_102	13	4	1.000	0	2.790	1072.381	0
US_104	93	4	0.914	0	2.135	1167.122	0
US_105	19	4	1.000	0	1.969	1126.073	0
US_108	3	4	1.000	0	2.607	1083.753	0
US_111	25	4	0.960	0	2.494	1001.687	0
US_114	22	4	0.773	0	1.829	1069.246	0
US_116	6	4	0.667	0	2.053	959.475	0
US_117	27	4	0.963	0	2.229	1017.507	0
US_118	16	4	0.938	0	2.795	1015.198	0
US_133	1	2	1.000	0	3.138	726.372	0
US_134	41	4	0.927	0	2.292	1108.676	0
US_136	33	4	1.000	0	2.151	1037.523	0
US_137	15	4	0.733	0	2.292	853.696	0
US_144	18	4	0.722	0	2.495	919.143	0
US_148	1	4	1.000	0	2.347	1103.106	0
US_149	15	4	1.000	0	1.479	1072.979	0
US_150	48	4	0.958	0	2.544	1081.437	0
US_151	10	4	1.000	0	0.819	1013.277	1
US_157	3	4	1.000	0	2.919	1140.657	0
US_158	1	4	1.000	0	2.912	798.142	0
US_161	4	2	1.000	0	3.149	733.763	0
US_166	2	2	1.000	0	2.967	667.354	0
US_173	1	2	1.000	0	3.015	335.757	0
US_180	1	4	1.000	0	3.476	769.721	0
US_183	5	4	1.000	0	2.822	1069.466	0
US_189	2	2	0.500	0	3.273	924.148	0
US_197	3	1	0.667	0	3.487	713.222	0
US_201	1	4	1.000	0	3.726	803.981	0
US_204	4	1	0.750	0	2.572	478.548	1
US_209	1	4	1.000	0	3.654	803.362	0
US_21	2	1	0.500	0	1.000	511.714	0
US_210	41	1	1.000	0	0.377	648.785	0
US_211	1	3	1.000	0	3.658	733.277	0
US_215	1	2	1.000	0	3.986	722.358	0
US_220	1	2	1.000	0	3.907	708.279	0
US_258	8	2	0.875	0	3.082	744.127	0
US_276	1	1	1.000	0	3.366	188.303	1
US_287	25	1	0.480	0	1.306	447.880	0
US_296	10	1	0.700	0	1.490	495.488	0
US_297	17	1	0.765	0	1.126	544.412	0
US_299	1	1	1.000	0	2.761	666.386	0
US_302	2	3	1.000	0	1.000	74.121	0
US_303	1	1	1.000	0	2.732	665.893	0
US_304	3	3	0.667	0	3.186	270.708	3
US_306	5	1	0.600	0	1.540	369.114	0
US_307	30	1	0.633	0	1.174	473.501	0
US_308	4	1	0.750	0	1.931	396.629	0
US_315	2	1	0.500	0	2.294	501.275	0
US_316	2	1	0.500	0	3.126	485.352	1

context	n_finds	dominant_phase	purity	mean_entropy	mean_energy	mean_local_sei	n_intrusions
US_319	6	1	0.500	0	1.098	542.349	0
US_324	1	3	1.000	0	3.269	276.655	1
US_72	5	4	0.800	0	3.553	894.390	0
US_76	11	1	0.455	0	1.106	436.797	0
US_89	1	2	1.000	0	3.631	702.375	0

Units with purity > 90% indicate consistent phase assignment within the context. Lower purity and higher mean entropy correspond to disturbed contexts where bioturbation or construction cuts have introduced material from adjacent phases.

```
n_pure <- sum(us_tab$purity >= 0.9)
cat(n_pure, "out of", nrow(us_tab), "units have purity >= 90%\n")
#> 33 out of 54 units have purity >= 90%
```

0.9 Phase Transition Matrix

Shows how often finds from phase i sit directly above finds from phase j , revealing spatial adjacency between phases:

```
tm <- phase_transition_matrix(fit)
print(tm)
#>      phase1 phase2 phase3 phase4
#> phase1      81      23      12       1
#> phase2      16      20       8      38
#> phase3      17       4       3       1
#> phase4       3      35       1     351
```

High off-diagonal values indicate zones of contact between phases. This is expected at the boundary between successive occupation periods.

0.10 Model Validation

When true phase assignments are known (e.g., from simulation), palimpsestr provides validation metrics. We demonstrate with the simulated `demo_moderate` dataset:

```
data(demo_moderate)
fit_val <- fit_sef(demo_moderate, k = 3, tafonomy = "taf_score", context = "context", seed = 42)
ari <- adjusted_rand_index(fit_val, demo_moderate$true_phase)
cat("Adjusted Rand Index:", round(ari, 3), "\n")
#> Adjusted Rand Index: 1
confusion_matrix(fit_val, demo_moderate$true_phase)
#>      true
#> estimated  2  1  3
#>      1 67  0  0
#>      2  0 67  0
#>      3  0  0 66
```

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_confusion(fit_val, demo_moderate$true_phase)
}
```

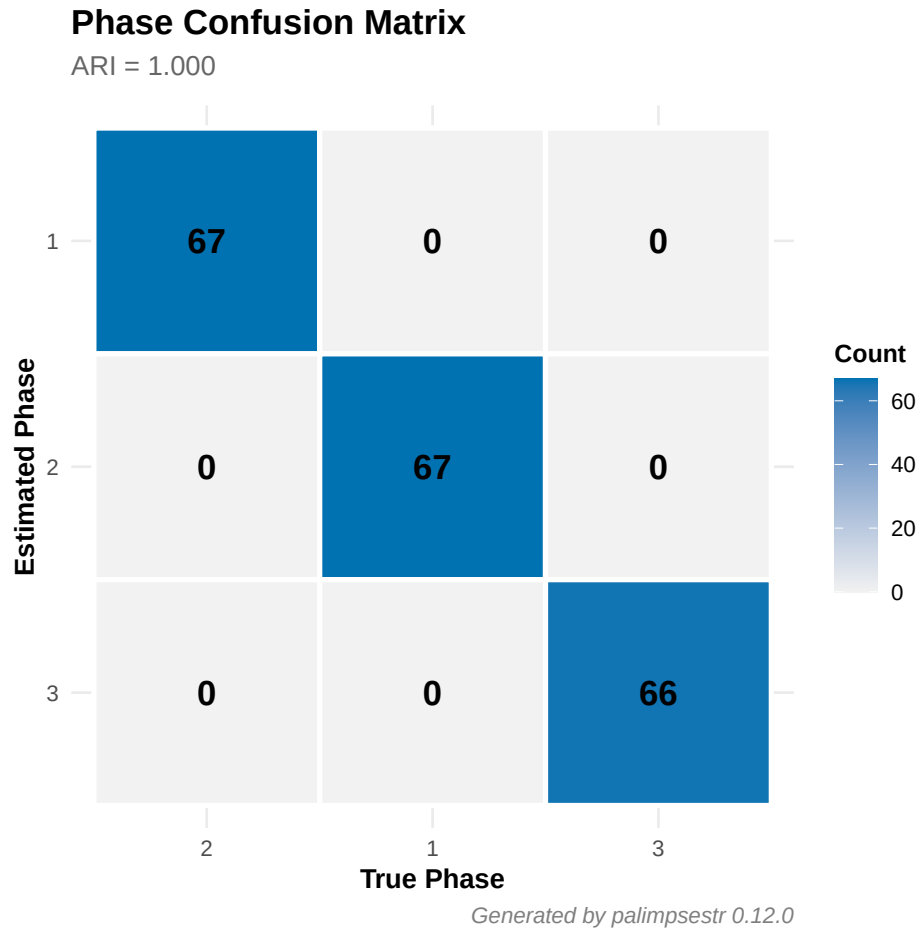


Figure 8: Confusion matrix heatmap on simulated data. Strong diagonal = correct phase recovery.

For the real Poggio Gramignano data, no ground truth is available. Validation relies on archaeological coherence: the model should recover phases consistent with the site periodisation, and flag known disturbed contexts (levelling fills, backfills) as mixed.

0.11 Sensitivity Analysis of SEI Weights

The SEI combines four evidence domains (spatial, vertical, temporal, cultural) with configurable weights. To assess how weight choices affect the results, we compare three configurations:

```
configs <- list(
  equal = c(ws = 1, wz = 1, wt = 1, wc = 1),
  spatial = c(ws = 2, wz = 1, wt = 0.5, wc = 0.5),
  temporal = c(ws = 0.5, wz = 0.5, wt = 2, wc = 1)
)
```

```

sens <- do.call(rbind, lapply(names(configs), function(nm) {
  w <- configs[[nm]]
  f <- fit_sef(villa_romana, k = 4, weights = w, seed = 42,
    tafonomy = "taf_score", context = "context")
  di <- detect_intrusions(f)
  data.frame(
    config = nm,
    pdi = pdi(f),
    mean_entropy = mean(f$entropy, na.rm = TRUE),
    n_intrusions = sum(di$intrusion_prob > 0.5),
    stringsAsFactors = FALSE
  )
}))
knitr::kable(sens, digits = 4,
  caption = "Sensitivity of PDI, mean entropy, and intrusion count to weight configuration.")

```

Table 3: Sensitivity of PDI, mean entropy, and intrusion count to weight configuration.

config	pdi	mean_entropy	n_intrusions
equal	1	0	12
spatial	1	0	16
temporal	1	0	13

The results show that the model is reasonably robust to weight variation, though emphasising the most informative domain for a given site can improve performance. The `optimize_weights()` function provides a data-driven approach to weight selection via cross-validated log-likelihood.

0.12 Bootstrap Confidence Intervals

Uncertainty on key statistics can be quantified via bootstrap resampling:

```

bs <- bootstrap_sef(fit, n_boot = 50, verbose = FALSE)
knitr::kable(bs, digits = 4,
  caption = "Bootstrap confidence intervals (50 replicates) for key model statistics.")

```

Table 4: Bootstrap confidence intervals (50 replicates) for key model statistics.

	statistic	estimate	lower	upper	se
pdi		1.0000	0.9976	1.0000	0.0007
mean_entropy		0.0000	0.0000	0.0034	0.0009
mean_energy		2.0053	1.9190	2.1284	0.0581
loglik		52469.2425	35066.1254	51194.5405	4385.6576

```

if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_bootstrap(bs)
}

```

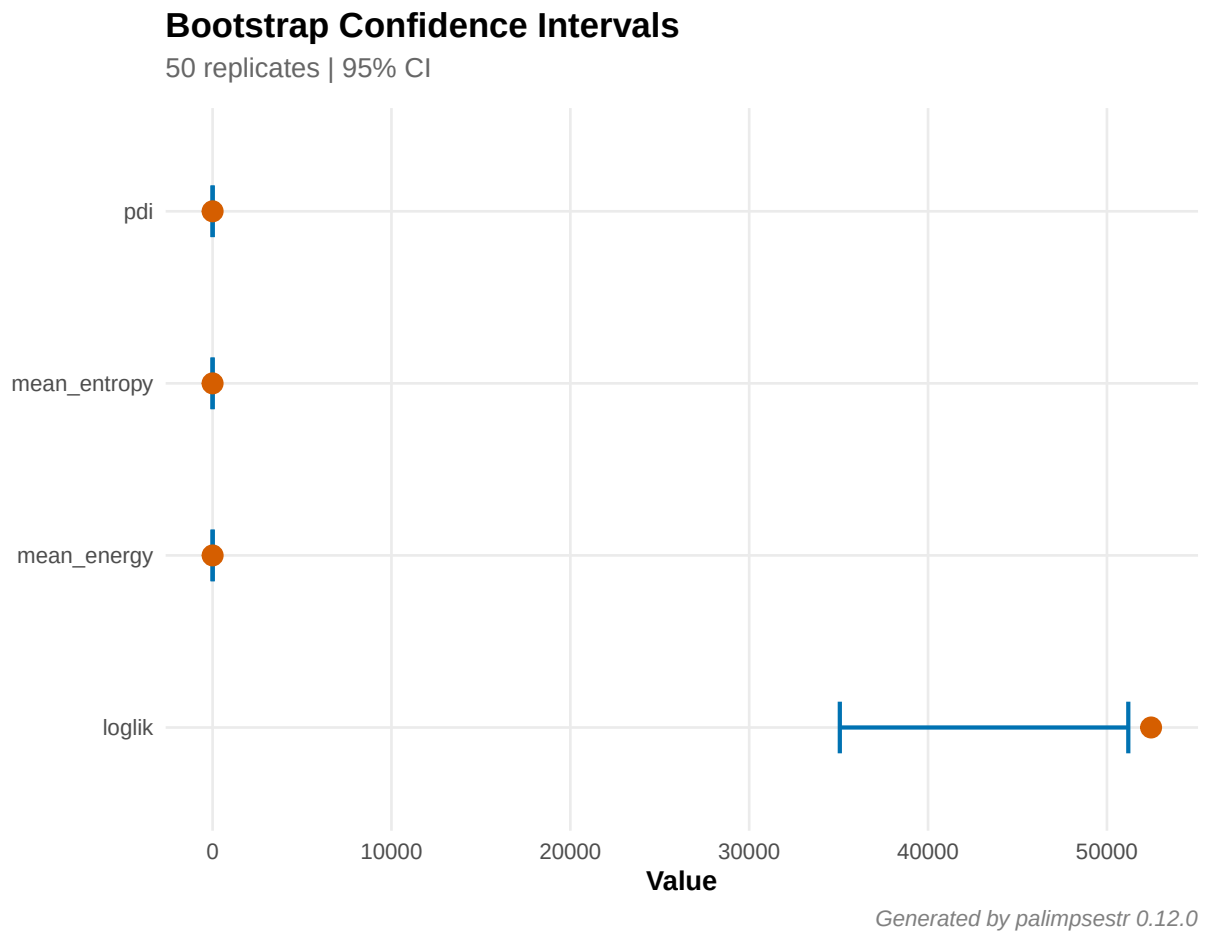


Figure 9: Bootstrap confidence intervals for PDI, mean entropy, mean energy, log-likelihood, and ARI.

0.13 Harris Matrix Integration

palimpsestr can incorporate stratigraphic information from a Harris Matrix as a penalty during model fitting. The `harris_from_contexts()` function auto-generates a penalty matrix from the mean depth of finds in each context. Alternatively, `read_harris()` imports an external Harris Matrix from a CSV edge list.

```
H <- harris_from_contexts(villa_romana, z_col = "z", context_col = "context")
cat("Harris penalty matrix:", nrow(H), "x", ncol(H), "\n")
#> Harris penalty matrix: 615 x 615
```

```
fit_h <- fit_sef(villa_romana, k = 4, harris = H,
                tafonomy = "taf_score", context = "context",
                n_init = 5, seed = 42)
cat("PDI without Harris:", round(pdi(fit), 4), "\n")
#> PDI without Harris: 1
cat("PDI with Harris: ", round(pdi(fit_h), 4), "\n")
#> PDI with Harris: 1
di_no <- detect_intrusions(fit)
di_h <- detect_intrusions(fit_h)
cat("Intrusions without Harris:", sum(di_no$intrusion_prob > 0.5), "\n")
#> Intrusions without Harris: 8
cat("Intrusions with Harris: ", sum(di_h$intrusion_prob > 0.5), "\n")
#> Intrusions with Harris: 8
```

0.13.1 Validate phases against stratigraphy

```
validate_phases_harris(fit)
#>      upper lower upper_phase lower_phase consistent
#> 1  US_302 US_303           3           1      FALSE
#> 2  US_303 US_299           1           1       TRUE
#> 3  US_299 US_304           1           3       TRUE
#> 4  US_304 US_296           3           1      FALSE
#> 5  US_296 US_297           1           1       TRUE
#> 6  US_297 US_308           1           1       TRUE
#> 7  US_308 US_287           1           1       TRUE
#> 8  US_287 US_306           1           1       TRUE
#> 9  US_306 US_316           1           1       TRUE
#> 10 US_316 US_307           1           1       TRUE
#> 11 US_307 US_324           1           3       TRUE
#> 12 US_324 US_315           3           1      FALSE
#> 13 US_315 US_76            1           1       TRUE
#> 14  US_76 US_204           1           1       TRUE
#> 15 US_204 US_319           1           1       TRUE
#> 16 US_319 US_276           1           1       TRUE
#> 17 US_276 US_151           1           4       TRUE
#> 18 US_151 US_210           4           1      FALSE
#> 19 US_210 US_173           1           2       TRUE
#> 20 US_173 US_136           2           4       TRUE
#> 21 US_136 US_166           4           2      FALSE
#> 22 US_166 US_102           2           4       TRUE
#> 23 US_102 US_137           4           4       TRUE
#> 24 US_137 US_108           4           4       TRUE
```

```

#> 25 US_108 US_118      4      4      TRUE
#> 26 US_118 US_144      4      4      TRUE
#> 27 US_144 US_89       4      2     FALSE
#> 28 US_89  US_161      2      2      TRUE
#> 29 US_161 US_158      2      4      TRUE
#> 30 US_158 US_157      4      4      TRUE
#> 31 US_157 US_114      4      4      TRUE
#> 32 US_114 US_104      4      4      TRUE
#> 33 US_104 US_116      4      4      TRUE
#> 34 US_116 US_209      4      4      TRUE
#> 35 US_209 US_211      4      3     FALSE
#> 36 US_211 US_201      3      4      TRUE
#> 37 US_201 US_21       4      1     FALSE
#> 38 US_21  US_134      1      4      TRUE
#> 39 US_134 US_189      4      2     FALSE
#> 40 US_189 US_197      2      1     FALSE
#> 41 US_197 US_215      1      2      TRUE
#> 42 US_215 US_220      2      2      TRUE
#> 43 US_220 US_258      2      2      TRUE
#> 44 US_258 US_180      2      4      TRUE
#> 45 US_180 US_105      4      4      TRUE
#> 46 US_105 US_133      4      2     FALSE
#> 47 US_133 US_149      2      4      TRUE
#> 48 US_149 US_148      4      4      TRUE
#> 49 US_148 US_72       4      4      TRUE
#> 50 US_72  US_111      4      4      TRUE
#> 51 US_111 US_150      4      4      TRUE
#> 52 US_150 US_117      4      4      TRUE
#> 53 US_117 US_183      4      4      TRUE

```

The **consistent** column flags whether the dominant phase ordering respects stratigraphic depth. **FALSE** indicates an inversion that may warrant investigation.

0.14 Interpretive Report

The **report_sef()** function generates a plain-language summary of the analysis, suitable for inclusion in excavation reports. It is available in English and Italian.

```

report_sef(fit, lang = "en")
#> # SEF Analysis Report
#>
#> ## Overview
#>
#> The Stratigraphic Entanglement Field model with **K = 4 phases** was fitted to a dataset of **615 fi
#>
#> The Palimpsest Dissolution Index (PDI) is **1.000**, indicating **excellent phase separation**. Mean
#>
#> Model diagnostics: LogLik = 52469.2, BIC = -103789.0.
#>
#> ## Phase Composition
#>
#> ### Phase 1 (117 finds, 19.0%)

```

```

#>
#> - **Dating**: average 69 CE (range: -650 to 550)
#> - **Material classes**: Materiale Da Costruzione (1), Reperti Archeozoologici (7), Reperto Anforaceo (1)
#> - **Mean entropy**: 0.0000; **Mean energy**: 1.037
#> - **Contexts**: US_197, US_204, US_21, US_210, US_276, US_287, ... (17 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 2 (82 finds, 13.3%)
#>
#> - **Dating**: average 413 CE (range: 138 to 550)
#> - **Material classes**: Materiale Da Costruzione (36), Reperto Anforaceo (11), Reperto Numismatico (1)
#> - **Mean entropy**: 0.0000; **Mean energy**: 2.436
#> - **Contexts**: US_104, US_111, US_114, US_116, US_117, US_118, ... (32 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 3 (25 finds, 4.1%)
#>
#> - **Dating**: average 367 CE (range: 180 to 450)
#> - **Material classes**: Campionature (5), Intonaco Dipinto (1), Materiale Da Costruzione (1), Materiale Numismatico (1)
#> - **Mean entropy**: 0.0000; **Mean energy**: 1.748
#> - **Contexts**: US_211, US_287, US_297, US_302, US_304, US_306, ... (10 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 4 (391 finds, 63.6%)
#>
#> - **Dating**: average 438 CE (range: 425 to 450)
#> - **Material classes**: Reperto Anforaceo (63), Reperto Ceramico (328)
#> - **Mean entropy**: 0.0000; **Mean energy**: 2.221
#> - **Contexts**: US_102, US_104, US_105, US_108, US_111, US_114, ... (26 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ## Intrusion Detection
#>
#> **8 finds** (1.3%) have an intrusion probability above 0.5, flagging them as potentially displaced or
#>
#> Top suspects:
#>
#> | ID | Context | Intrusion Prob. |
#> |---|---|---|
#> | VRPG_0402 | US_151 | 0.588 |
#> | VRPG_0489 | US_276 | 0.573 |
#> | VRPG_0431 | US_204 | 0.559 |
#> | VRPG_0590 | US_316 | 0.539 |
#> | VRPG_0598 | US_324 | 0.539 |
#> | VRPG_0548 | US_304 | 0.538 |
#> | VRPG_0547 | US_304 | 0.537 |
#> | VRPG_0546 | US_304 | 0.525 |
#>
#> ## Stratigraphic Unit Purity
#>
#> Of **54 stratigraphic units**: **33 pure** (>90%), **9 mixed** (70-90%), **12 palimpsest** (<70%).
#>
#> Units classified as palimpsest require special attention, as material from multiple depositional epi

```



```

#>
#> ## Recommendations
#>
#> The deposit shows good phase separation. The model can be used with confidence for phase attribution
#>
#> ---
#> *Report generated by palimpsestr 0.12.0*

```

```

report_sef(fit, lang = "it")
#> # Rapporto Analisi SEF
#>
#> ## Panoramica
#>
#> Il modello Stratigraphic Entanglement Field con K = 4 fasi e' stato applicato a un dataset di 117 reperti
#>
#> Il Palimpsest Dissolution Index (PDI) e' 1.000, indicando una separazione eccellente delle fasi
#>
#> Diagnostiche: LogLik = 52469.2, BIC = -103789.0.
#>
#> ## Composizione delle Fasi
#>
#> ### Fase 1 (117 reperti, 19.0%)
#>
#> - Datazione media: 69 d.C.
#> - Classi materiali: Materiale Da Costruzione (1), Reperti Archeozoologici (7), Reperto Anforaceo (9)
#> - Entropia media: 0.0000; Energia media: 1.037
#> - US: US_197, US_204, US_21, US_210, US_276, US_287, ... (17 totali)
#>
#> ### Fase 2 (82 reperti, 13.3%)
#>
#> - Datazione media: 413 d.C.
#> - Classi materiali: Materiale Da Costruzione (36), Reperto Anforaceo (11), Reperto Numismatico (35)
#> - Entropia media: 0.0000; Energia media: 2.436
#> - US: US_104, US_111, US_114, US_116, US_117, US_118, ... (32 totali)
#>
#> ### Fase 3 (25 reperti, 4.1%)
#>
#> - Datazione media: 367 d.C.
#> - Classi materiali: Campionature (5), Intonaco Dipinto (1), Materiale Da Costruzione (1), Materiale Anforaceo (19)
#> - Entropia media: 0.0000; Energia media: 1.748
#> - US: US_211, US_287, US_297, US_302, US_304, US_306, ... (10 totali)
#>
#> ### Fase 4 (391 reperti, 63.6%)
#>
#> - Datazione media: 438 d.C.
#> - Classi materiali: Reperto Anforaceo (63), Reperto Ceramico (328)
#> - Entropia media: 0.0000; Energia media: 2.221
#> - US: US_102, US_104, US_105, US_108, US_111, US_114, ... (26 totali)
#>
#> ## Intrusioni
#>
#> 8 reperti (1.3%) presentano probabilita' di intrusione superiore a 0.5.
#>

```

```
#> ## Purezza delle US
#>
#> Su **54 US**: **33 pure** (>90%), **9 miste** (70-90%), **12 palinsesto** (<70%).
#>
#> ---
#> *Rapporto generato da palimpsestr 0.12.0*
```

0.15 Structured Exports

All results can be exported to CSV for integration with external databases and GIS:

```
export_results(fit, dir = "results/", format = "csv", prefix = "villa_romana")
```

This creates 4 files:

- `villa_romana_phases.csv` — phase assignments and diagnostics per find
- `villa_romana_intrusions.csv` — intrusion probabilities
- `villa_romana_us_summary.csv` — per-US aggregated statistics
- `villa_romana_model_summary.csv` — model-level metrics (n, k, PDI, BIC, etc.)

0.16 GIS Export

For integration with GIS workflows, `palimpsestr` can export results as `sf` spatial objects:

```
if (requireNamespace("sf", quietly = TRUE)) {
  sf_pts <- as_sf_phase(fit, crs = 32632L, dims = "XYZ")
  sf_links <- as_sf_links(fit, quantile_threshold = 0.90, crs = 32632L)
  sf::st_write(sf_pts, "villa_romana.gpkg", layer = "phases")
  sf::st_write(sf_links, "villa_romana.gpkg", layer = "sei_links")
}
```

0.17 Interactive Visualization

When `plotly` is available, any `gg_*` plot can be converted to an interactive visualization using `as_plotly()`:

```
as_plotly(gg_phasefield(fit))
as_plotly(gg_intrusions(fit))
```

0.18 Comparison with Traditional Approaches

Aspect	Harris Matrix	k-means	palimpsestr (SEF)
Evidence domains	Stratigraphy only	Spatial only	Spatial + vertical + temporal + cultural
Assignment type	Deterministic	Hard	Probabilistic (soft)
Mixing detection	No	No	Yes (entropy, ESE)

Aspect	Harris Matrix	k-means	palimpsestr (SEF)
Intrusion detection	Manual	No	Automatic
Scalability	Manual	Good	Good
Interpretability	High	Low	High (PDI, reports)

The SEF framework does not replace the Harris Matrix. Rather, it extends it. The Harris Matrix remains indispensable for documenting observed physical relationships between depositional units. What palimpsestr adds is a quantitative, probabilistic layer of analysis that operates on find-level data within those units — precisely the scale at which palimpsest formation is most problematic.

0.19 Model Limitations and Methodological Notes

0.19.1 Diagonal Covariance Assumption

The EM uses diagonal Gaussians, assuming conditional independence between features within each phase. This is a deliberate parsimony choice: for p features and k phases, diagonal covariance requires $k(2p + 1) - 1$ parameters versus $k(p(p + 3)/2 + 1) - 1$ for full covariance. With typical archaeological datasets ($n = 50$ – 500 , $p = 7$ – 10), full covariance risks overfitting.

0.19.2 SEI Component Normalisation

Each SEI component is normalised to $[0, 1]$ by dividing by its within-dataset maximum. This makes weights interpretable but means **absolute SEI values are not comparable across datasets**. Cross-site comparisons should use derived statistics (PDI, mean entropy) which are scale-invariant.

0.19.3 PDI Interpretation

The PDI is a descriptive measure, not a formal test statistic. The interpretive guideline ($PDI > 0.7 =$ well-resolved) is empirically derived from simulation studies and should not be treated as a formal decision boundary. Bootstrap confidence intervals provide uncertainty quantification.

0.19.4 Intrusion Detection

The intrusion probability score is a heuristic combining rescaled entropy, ESE, and inverse local SEI. The 0.5 threshold is an exploratory guideline. Users should examine flagged observations individually using domain knowledge.

0.19.5 Phase Labelling

Phase numbers are arbitrary in mixture models (label switching). Use `reorder_phases()` to ensure phase 1 = deepest (oldest) stratum. The bootstrap procedure applies phase reordering to each replicate.

0.19.6 Cross-Validation

The `cv_sef()` and `optimize_weights()` functions standardise test data using the training set’s center and scale, ensuring consistent feature scales across folds.

0.19.7 Scalability

The SEI and ESE matrices are $O(n^2)$ in memory. For datasets exceeding ~2000 finds, use `sei_sparse()` or the `max_dist` parameter in `sei_matrix()` to limit computation to spatial neighbours.

0.20 Interactive Shiny Dashboard

palimpsestr includes a built-in Shiny application for interactive analysis. The dashboard supports CSV upload, SQLite and PostgreSQL connections, and provides all diagnostic plots, tables, and maps in a point-and-click interface.

```
launch_app()
```

The app has six tabs: Data Input (CSV/SQLite/PostgreSQL with column mapping), Analysis (parameter selection and model fitting), Results (interactive plots via plotly), Tables (phase assignments, intrusions, US purity), Maps (overlay on excavation plan geometries), and Report (interpretive report generation and CSV/ZIP download).

0.21 Advanced Feature Space Options

Version 0.12.0 introduces five optional improvements to the feature space that address limitations observed in real-data testing (Poggio Gramignano villa, Solarolo terramare).

0.21.1 Chronological Precision Weighting

Finds with precise dating (narrow date range) should contribute more to temporal clustering than finds with generic dating. When `chrono_precision = TRUE`, the feature `1/tspan` is added, giving higher weight to precisely dated finds.

```
# Finds dated "BM2b" (50-year range) weight more than "BM" (300-year range)
fit <- fit_sef(data, k = 3, chrono_precision = TRUE)
```

0.21.2 Residuality Detection

When a find's dating is inconsistent with its stratigraphic context (e.g., Late Antique ceramic in a Pre-Roman pit fill), the `residuality` feature captures this mismatch. Requires `context`.

```
# Adds abs(tmids - tmids_context_mean) / max_range as a feature
fit <- fit_sef(data, k = 4, context = "context", residuality = TRUE)
```

0.21.3 Class Scaling

By default, one-hot encoded class columns (0/1) are not scaled, causing them to dominate the clustering when many classes are present. `class_scale = TRUE` multiplies one-hot columns by $1/\sqrt{n_{classes}}$, putting their total energy on par with a single numeric feature.

```
fit <- fit_sef(data, k = 4, class_scale = TRUE)
```

0.21.4 Taphonomic Score as Feature

In addition to using `taf_score` as a weight in the EM algorithm, it can be added as an explicit feature dimension, so that finds with similar taphonomic conditions cluster together.

```
fit <- fit_sef(data, k = 4, tafonomy = "taf_score", taf_as_feature = TRUE)
```

0.21.5 Sub-class Encoding

When finer typological distinctions are available (e.g., “ceramic_impasto” vs “ceramic_sigillata” instead of just “ceramic”), the `subclass` parameter uses them for one-hot encoding without changing the main `class` column.

```
fit <- fit_sef(data, k = 4, subclass = "subtype")
```

0.21.6 Using All Five Together

```
fit <- fit_sef(data, k = 4,
  tafonomy = "taf_score", context = "context",
  chrono_precision = TRUE,
  taf_as_feature = TRUE,
  residuality = TRUE,
  class_scale = TRUE,
  subclass = "subtype")
```

0.22 Conclusion

The **palimpsestr** package offers a unified probabilistic framework for decomposing archaeological palimpsests, addressing a long-standing gap between qualitative stratigraphic reasoning and quantitative spatial analysis. By jointly modelling spatial proximity, vertical distribution, chronological overlap, and cultural similarity, the Stratigraphic Entanglement Field approach produces soft phase assignments that acknowledge — rather than suppress — the inherent uncertainty of mixed deposits. The three diagnostic statistics (SEI, ESE, PDI) provide interpretable, actionable summaries at the find, deposit, and assemblage levels respectively.

The package is available on GitHub at <https://github.com/enzococca/palimpsestr>.

```
# install.packages("remotes")
remotes::install_github("enzococca/palimpsestr")
```